

Consensus on Lie Groups in the Presence of Unreliable Information Networks

Research and Development Project

Anshu Arora

Under the guidance of
Prof. Ravi Banavar

1 Introduction

Consensus is a fundamental problem in distributed systems, where a network of agents seeks to reach agreement on a common state. It has broad applications ranging from modeling social dynamics to distributed optimization and swarm coordination. While consensus in Euclidean spaces is well understood—including settings with intermittent, unreliable, or time-varying communication, many real-world systems do not evolve in flat vector spaces. Practical applications such as attitude synchronization, formation control, and robotic coordination naturally evolve on Lie groups. However, the lack of global linear structure complicates classical consensus analysis. In this work, we identify this gap and take steps toward bridging it.

2 Preliminaries

Let us begin this interesting problem by revising some key notions that will help us understand Lie Groups.

2.1 Groups

A **group** $(G, *)$ is a set equipped with a binary operation that satisfies:

- **Associativity:**

$$g_1 * (g_2 * g_3) = (g_1 * g_2) * g_3, \quad \forall g_1, g_2, g_3 \in G.$$

- **Identity element:** There exists an element $e \in G$ such that

$$g * e = e * g = g, \quad \forall g \in G.$$

- **Inverse element:** For every $g \in G$, there exists an inverse $g^{-1} \in G$ such that

$$g * g^{-1} = g^{-1} * g = e.$$

2.2 Smooth Manifolds

A **smooth manifold** M is a topological space that locally resembles Euclidean space and admits a collection of coordinate charts. This structure allows one to perform calculus on M .

A map $\phi : M_1 \rightarrow M_2$ between smooth manifolds is called a **diffeomorphism** if:

1. ϕ is smooth,
2. its inverse ϕ^{-1} exists and is smooth.

2.3 Lie Groups

A **Lie group** is a mathematical object that is simultaneously a group and a smooth manifold. Specifically, it consists of:

- a **group structure** with multiplication $*$ and inverse $(\cdot)^{-1}$, and
- a **smooth manifold structure** that allows for differentiable parametrizations.

The defining smoothness property of a Lie group requires that the maps

$$(g_1, g_2) \mapsto g_1 * g_2, \quad g \mapsto g^{-1}$$

are smooth with respect to the manifold structure.

Lie groups arise naturally as spaces of continuous symmetries, linking algebraic group operations with tools from differential geometry. They naturally model the dynamics of many systems, which is why we are interested in them!

3 Problem Formulation

We study the problem of consensus of agents on Lie groups in the presence of time-varying communication between agents. The communication structure among agents is modelled as a weighted, undirected graph, where the connectivity may be partially or entirely lost for finite durations of time.

Consider a group of N agents, denoted $\{g_i\}_{i=1}^N$, each evolving on a Lie group G . The agents are connected via a time-varying, undirected, weighted communication graph with weights $a_{ij}(t) \geq 0$. This interaction topology is governed by a *switching function* $\sigma : \mathbb{R}_{\geq 0} \rightarrow \mathcal{P}$, where $\mathcal{P}$ is the set of possible communication graphs. At any time t , $\sigma(t)$ specifies the set of edges that are active ($a_{ij} > 0$), as well as their weights. We would like to use the kinematic model of consensus, or the laplacian flow, given as

$$\text{First-order: } v_i = -k_P \mathbf{grad}^i V_{TOT},$$

$$\text{Second-order: } \nabla_{v_i} v_i = -k_P \mathbf{grad}^i V_{TOT} - k_D v_i.$$

1. Here v_i represents the left-translated velocity of the agent, which is $g_i^{-1} * \dot{g}_i$.
2. V_{TOT} is the weighted sum of all the potential energies, given as $\frac{1}{2} \sum_{ij} a_{ij} V_{ij}$.
3. V is a G-polar navigation function, so that minimizing this function leads to the unique minimum of potential on the manifold.
4. V_{ij} represents the 'potential' between the two agents, and it is formally $V(E_{ij}^L)$.
5. We note that E_{ij}^L is the left translated error, given as $g_j^{-1} * g_i$.

6. $\mathbf{grad}$ represents the left-translated gradient
7. $\nabla_{v_i} v_i$ represents the acceleration on the lie-group, corrected for the curvature in the manifold.

Intuitively, one should think of the following framework of kinematic consensus as having springs and damping objects (in the case of second-order consensus) between the agents, that try to bring all agents together. With this picture in mind, it is clear that if the graph is always connected, the equivalent imagined mechanical system will converge to a single point (imagine point agents and length-less springs and dampers).

Our goal is to study the evolution of the system and the convergence properties of consensus protocols under the dynamics induced by $\sigma(t)$. Primary questions of interest to us are:

1. How does intermittent or lossy connectivity affect the ability of the system to achieve consensus? For the first-order dynamics

$$v_i(t) = -k_P \mathbf{grad}^i V_{\text{TOT}}(t), \quad (1)$$

does there exist a switching signal $\sigma(t)$ such that

$$\lim_{t \rightarrow \infty} \|x_i(t) - x_j(t)\| = 0, \quad \forall i, j \in \{1, \dots, N\}, \quad (2)$$

even when $E_{\sigma(t)}$ is intermittently disconnected?

2. What is the domain of initial conditions from which consensus can still be achieved under such communication disruptions?

For a given class of switching signals Σ , determine the set

$$\mathcal{D} = \left\{ (x(0), v(0)) \mid \lim_{t \rightarrow \infty} \|x_i(t) - x_j(t)\| = 0 \quad \forall i, j \right\}. \quad (3)$$

3. Can we characterize the relationship between the *duration* of connectivity loss and the convergence guarantees (e.g., convergence time or error bounds)?

Suppose the network is disconnected over intervals of length at most $\tau_{\max}$. Find conditions on $\tau_{\max}$ such that

$$\exists T(\varepsilon, \tau_{\max}) < \infty : \quad \|x_i(t) - x_j(t)\| < \varepsilon, \quad \forall t \geq T, \quad (4)$$

for all i, j and $\varepsilon > 0$.

4. What is the maximum number of edges that can simultaneously lose connectivity without affecting consensus guarantees?

Let $\delta(t)$ denote the number of missing edges at time t . Determine the maximal integer δ^* such that consensus is guaranteed if

$$\delta(t) \leq \delta^*, \quad \forall t \geq 0. \quad (5)$$

4 Solutions on the Euclidean Space

The traces of the first solution to this problem, with connections that vary over time, were presented in [2]. We go through the mathematical tools and assumptions that they used to derive their key result: that the connection topology must form a connected graph over uniformly bounded time intervals, in order for consensus to occur. We focus on the leaderless setting

Model (discrete-time Vicsek variant): Consider N agents with headings $\theta_i(t) \in [-\pi, \pi)$ at discrete times $t \in \mathbb{Z}_{\geq 0}$. Let $\mathcal{N}_i(t)$ denote the neighborhood (agents within sensing radius) of agent i at time t . The **Vicsek update** is

$$\begin{aligned} \theta_i(t+1) &= \bar{\theta}_i(t) \\ \bar{\theta}_i(t) &:= \frac{1}{1 + |\mathcal{N}_i(t)|} \left(\theta_i(t) + \sum_{j \in \mathcal{N}_i(t)} \theta_j(t) \right). \end{aligned} \quad (6)$$

Writing the state vector $\theta(t) = [\theta_1(t), \dots, \theta_n(t)]^\top$, the update becomes

$$\theta(t+1) = F_{\sigma(t)} \theta(t), \quad (7)$$

where

$$F_p = (I + D_p)^{-1} (A_p + I)$$

where A_p is the adjacency matrix of graph $\mathcal{G}_p$ and D_p is the diagonal matrix whose i -th diagonal element is the valence of vertex i within the graph, and $\sigma(t)$ indexes the (time-varying) undirected neighbor graph.

Main mathematical analysis techniques:

- **Products of stochastic matrices / Wolfowitz’s theorem.** The paper reduces convergence to studying finite products of row-stochastic, nonnegative matrices with positive diagonals (aperiodic and ergodic matrix) and then applies the Wolfowitz about finite products of ergodic matrices: if every finite product drawn from a finite set of ergodic matrices is ergodic, then products converge to rank-one matrices (consensus).

Main results and insights:

- If every graph encountered at every time is connected (i.e., $\sigma(t) \in \mathcal{G}_{\text{conn}}$ for all t), then $\theta_i(t) \rightarrow \theta^*$ (all headings agree) as $t \rightarrow \infty$. Convergence to a common value follows since each F_k is ergodic/primitive in this case, hence so is their product. Moreover, in the case the graph is always fully connected, we can even visualize the consensus happening, by modelling springs and damping blocks (if second order flow) between the agents.
- If there exists an **infinite sequence of contiguous, bounded time intervals** $[t_\ell, t_\ell + \tau_\ell]$ (with uniformly bounded $\sup_\ell \tau_\ell$) such that on each interval the collection of graphs encountered is jointly connected, then consensus occurs: $\theta(t) \rightarrow \theta^* \mathbf{1}$

Limitations and caveats:

- **Dependence on topology assumptions** Convergence requires sufficiently frequent joint connectivity; The conditions are sufficient but not necessary—the exact weakest connectivity conditions remain open.
- **Noise and modeling gaps** The analysis is deterministic and ignores measurement noise, sensing errors, collision avoidance, and exact motion/kinematics — the authors note these are important for practical robot teams and that the deterministic results suggest focusing on connectivity when adding noise. We will not look into this limitation for the moment.

To overcome the first caveat, work done by Ren and Beard [4], tried to loosen the assumption required for consensus: that the union of the interaction graph should form a connected graph in finite intervals of time. They

show that a spanning tree requirement is sufficient for consensus, effectively reducing the requirement to half of the communication links presented in [2]. Also, they generalized the vicsek model of consensus, to any weighted average of the neighbouring heading directions,

However, in order to extend and interpret and extend these results on Lie Groups, there are some issues we need to tackle:

- Both these approaches are based on a linear model, where we are cleanly able to write the updates in heading direction in a matrix form
- As we have seen, work done on consensus on Lie Groups requires the kinematic model of consensus, which is non-linear

Our first step is to try to generalize such results on requirements of interaction topology to models of consensus that are not linear, and this very problem is tackled in Luc Moreaus work [3].

Moving to a general framework: To guarantee consensus on the Euclidean space, with the control given as $x(t+1) = f(t, x(t))$ the following assumptions must hold.

Associated to each directed graph $(\mathcal{N}, \mathcal{A})$ with node set $\mathcal{N} = \{1, \dots, N\}$, each agent k and each state $x \in X^n$, there is a compact set $e_k(\mathcal{A})(x) \subset X$ satisfying

1. $f_k(t, x) \in e_k(\mathcal{A}(t))(x), \quad \forall t \in \mathbb{N}, \forall x \in X^n;$ (4)
2. $e_k(\mathcal{A})(x_1, \dots, x_n) = \{x_k\}$ whenever the states of agent k and agents $i \in \text{Neighbors}(k, \mathcal{A})$ are all equal;
3. $e_k(\mathcal{A})(x_1, \dots, x_n)$ is contained in the relative interior of the convex hull of the states of agent k and agents $i \in \text{Neighbors}(k, \mathcal{A})$ whenever the states of agent k and agents $i \in \text{Neighbors}(k, \mathcal{A})$ are not all equal;
4. $e_k(\mathcal{A})(x)$ depends continuously on x ; that is, the set-valued function $e_k(\mathcal{A}) : X^n \rightrightarrows X$ is continuous.

Essentially, what we can understand from the requirement of this assumption, is that as the agents progress in time, the convex hull of their error set should keep decreasing over sufficiently long time intervals. We can relate this to the insight in paper [2]. After time τ_ℓ , the hull will decrease, and hence we will eventually reach consensus.

Key takeaway from this paper: Once the abstraction is made to convex hulls, through simple reasoning machinery- and set-valued Lyapunov theory, the required proofs can be made. There are no graph-theoretic results employed, just visualization of the convex hull shrinking will help us!

The interesting part is that most of this paper relies on using convexity for all of its proofs on unidirectional and bidirectional communication. If we can just extend this result to the Laplacian flow on Lie groups we are done. However -how will we model this convexity on the Lie Group?

Let us first look into the proof steps of how consensus on Lie Group is proven mathematically, to see where will require some extra thinking.

5 Consensus on Lie Groups

We will analyse the requirements for consensus and stability analysis presented in the work [1]. We begin by stating the following key theorems.

Gradient Flows on Riemannian Manifolds:

Theorem 1 (Gradient Descent Flow on Riemannian Manifolds) *Let $F : M \rightarrow \mathbb{R}$ be a smooth scalar function on a smooth Riemannian manifold M , and let $k_P, k_D > 0$. Consider the following dynamical systems:*

$$\dot{\gamma} = -k_P \text{grad}(F), \quad (8)$$

$$\nabla_{\dot{\gamma}} \dot{\gamma} = -k_P \text{grad}(F) - k_D \dot{\gamma}. \quad (9)$$

If all sub-level sets of F are compact (for instance, trivially so if M itself is compact), then for any critical point $c^ \in C := \{x \mid \text{grad}_x(F) = 0\}$, every integral curve $\gamma(t)$ of either system satisfies:*

- $\lim_{t \rightarrow \infty} \gamma(t) = c^*,$
- $\lim_{t \rightarrow \infty} (\gamma(t), \dot{\gamma}(t)) = (c^*, 0).$

Thus, solutions converge to critical points of F .

Interpretation: A smooth curve on a Riemannian manifold follows the negative gradient vector field and converges asymptotically to a critical point.

Invariant Sets and LaSalle's Principle:

Definition 5.1 (Invariant Set) *For a dynamical system $\dot{x} = f(x)$, a set $\mathcal{W}$ is invariant if*

$$x(0) \in \mathcal{W} \implies x(t) \in \mathcal{W}, \quad \forall t \geq 0.$$

Theorem 2 (LaSalle's Invariance Principle) *Let $V(x)$ be a continuously differentiable Lyapunov function such that $\dot{V}(x) \leq 0$ for all x . Then every trajectory converges to the largest invariant subset of*

$$\{x \mid \dot{V}(x) = 0\}.$$

From these two

Proposition 5.1 (Negative Gradient Systems) *For gradient systems of the form*

$$\dot{x} = -\nabla V(x),$$

one has

$$\dot{V}(x) = -\|\nabla V(x)\|^2 \leq 0,$$

and therefore $V(x)$ is non-increasing along system trajectories.

Proposition 5.2 (Consensus as the Invariant Set) *The largest invariant set satisfying $\nabla V(x) = 0$ corresponds to the equilibrium condition for consensus, i.e.,*

$$p_1 = p_2 = \dots = p_n = p^*.$$

Hence, by LaSalle's principle, and gradient descent flow of Riemannian manifolds, all agent trajectories converge to this consensus configuration. The Lie group considered is equipped with a bi-invariant metric [1], hence by definition the set is compact and we can apply these results. Hence, we have consensus on the Lie Group (when there is no constraint on the communication)

We now look into the stability analysis for the same.

Center Manifold Theory:

Theorem 3 (Center Manifold Emergence Theorem) *Let W be a smooth vector field on a Riemannian manifold G and let g^* be an equilibrium point with $W(g^*) = 0$. Suppose the Jacobian $J_{g^*}W$ has p eigenvalues equal to zero, and the remaining eigenvalues have strictly negative real parts. Let ν be the maximum real part among the non-zero eigenvalues. Then:*

- *There exists a locally unique p -dimensional invariant submanifold, the center manifold M_{cen} , passing through g^* and tangent to $\ker(J_{g^*}W)$.*
- *For sufficiently small neighbourhood N_{cen} , any integral curve beginning in N_{cen} converges exponentially to an invariant trajectory contained in $M_{\text{cen}} \cap N_{\text{cen}}$, with rate ν .*

Local Stability of Consensus:

Theorem 4 (Local Exponential Stability of Consensus) *For any connected symmetric interaction graph, the consensus equilibrium is locally exponentially stable for both first-order and second-order consensus dynamics.*

The consensus manifold is

$$M_C := \{(g, g, \dots, g) : g \in G\}.$$

Consensus corresponds to the condition $\text{grad}_x(F) = 0$, and all agent states converge to a common equilibrium g^* .

Proof Sketch of Theorem:

1. The left Lie-block Jacobian of the consensus dynamics is shown to be negative semi-definite.
2. This is established by identifying a Laplacian-like structure in the Jacobian and using the standard factorization properties of graph Laplacians.
3. The manifold M_C is invariant: once agents reach consensus, they remain in consensus.
4. By the Center Manifold Theorem, all trajectories converge exponentially to M_C .

Thus, the consensus equilibrium is locally exponentially stable.

6 Future Works and Summary

All that remains for us now, is patching the two frameworks of consensus even in the presence of intermittent connections (on the Euclidean Space), and consensus on the Lie Group. The key link to perform this comes from the convex hull analysis present in [3]. However, we do not have a global notion of convexity on the Lie Group, so we will have to carefully define a mathematical objective that can be used as a proxy. While one may feel that pushing the agents to the Lie algebra may help resolve this problem, since it is a linear vector space, it actually doesn't carry the information of the relative configurations between the agents that we require! So even though we could perform this convexity analysis there, it would not lead to anything fruitful.

Here are some promising leads I could find for notions of convexity on Lie Groups, that I plan to pursue in the next semester-

- I first looked into gradient descent on Riemannian manifolds, and optimization on manifolds, however that uses geodesic convexity, which can not be meaningfully used in our setting.
- Kostant's convexity theorem - projection of every coadjoint orbit of a connected compact Lie group into the dual of a Cartan subalgebra is a convex set. I do not really understand this theorem and its terminology very well, so I will have to look into it further.
- Convex hull on a Riemannian manifold- Define a subset C of M to be convex if all shortest paths between any two points $x, y \in C$ are completely contained within C . For a finite set of points P on M , define the convex hull of P to be the intersection of all convex sets containing P . While this seems simple enough, most of the convenient properties of convex hulls on Euclidean spaces do not translate to this setting
- Other approaches could be defining the notion of "volume", or a similar high-dimensional analog, and see how that evolves with time.

7 Recommendations

From Prof. Twinkle, to go through the following works:

- **KTH: Prof. Karl H. Johansson:** Study of *uniform connectivity*, *essential connectivity*, and *weakly connected networks*, with emphasis on graph-theoretic conditions for coordination and convergence in networked systems.
- **Rendezvous on Unicycles: Daniel Liberzon:** Lyapunov-based analysis for rendezvous problems, focusing on the precise design of switching laws to guarantee desired collective behavior.
- **Switching and Graph Conditions:** Investigation of how switching strategies can be designed to preserve or enforce required graph-theoretic properties, and whether similar connectivity conditions are maintained across switching topologies.

She also pointed out that the name 'unreliable information network' is a little ambiguous, and we should make it more precise.

From Prof. Constantine: Go through the work done by : D Astolfi, M Giaccagli, Andreu Cecelia, Samuele Zoboli. Also, can you come up with simple analysis of maximum of a set not increasing with time, to get an alternate viewpoint of consensus?

References

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